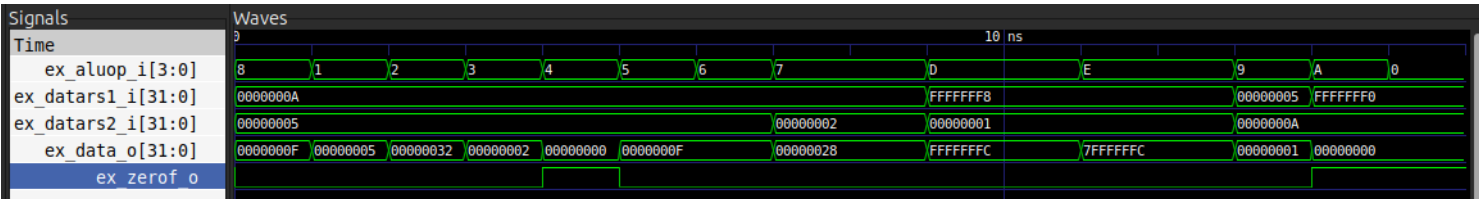


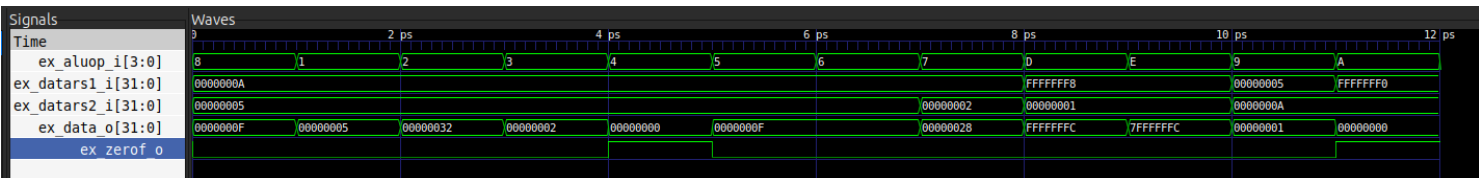
iverilog

```
manbenit@NimShuriken:~/Gitlab/uvm-exercices/testbenches/01_iverilog(main)$ iverilog -o intermedio ALU_tb.v ../models/alu_monociclo/*
manbenit@NimShuriken:~/Gitlab/uvm-exercices/testbenches/01_iverilog(main)$ vvp intermedio
VCD info: dumpfile dut_signals.vcd opened for output.
=== START TESTBENCH ===
Add:      10 +      5 =      15 (expected 15)
Sub:      10 -      5 =       5 (expected 5)
Prod:     10 *      5 =     50 (expected 50)
Div:      10 /      5 =       2 (expected 2)
AND: 000000000000000000000000000000001010 & 00000000000000000000000000000000101 = 00000000000000000000000000000000
XOR: 000000000000000000000000000000001010 ^ 00000000000000000000000000000000101 = 000000000000000000000000000000001111
OR: 000000000000000000000000000000001010 | 00000000000000000000000000000000101 = 000000000000000000000000000000001111
Shift Left 2:      10 << 2 =      40
Shift Left 2: 000000000000000000000000000000001010 << 2 = 00000000000000000000000000000000101000
Shift Right Arith: 4294967288 >>>      1 = 4294967292
Shift Right Arith: 1111111111111111111111111111000 >>> 000000000000000000000000000000000001 = 11111111111111111111111111111100
Shift Right Logical: 4294967288 >>      1 = 2147483644
Shift Right Logical: 1111111111111111111111111111000 >> 000000000000000000000000000000000001 = 01111111111111111111111111111100
SLT:      5 <      10 ?      1
SLTU: ffffffff0 < 0000000a ?      0
Default: output =      0 (expected 0)
=== END TESTBENCH ===
ALU_tb.v:97: $stop called at 16000 (lps)
** VVP Stop(0) **
** Flushing output streams.
** Current simulation time is 16000 ticks.
> exit
** Continue **
manbenit@NimShuriken:~/Gitlab/uvm-exercices/testbenches/01_iverilog(main)$
```



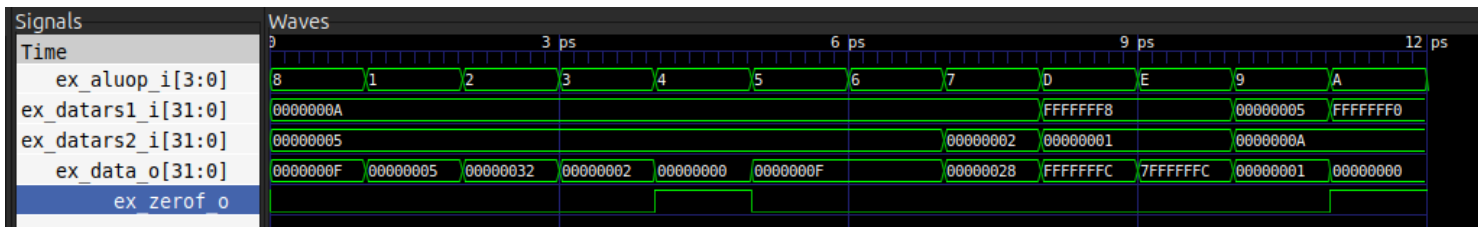
verilator

```
manbenit@NimShuriken:~/Gitlab/uvm-exercices/testbenches/02_verilator(main)$ ./obj_dir/alutb
=== START TESTBENCH ===
Add: 10 + 5 = 15 (expected 15)
Sub: 10 - 5 = 5 (expected 5)
Prod: 10 * 5 = 50 (expected 50)
Div: 10 / 5 = 2 (expected 2)
AND: 000000000000000000000000000000001010 & 00000000000000000000000000000000101 = 00000000000000000000000000000000
XOR: 000000000000000000000000000000001010 ^ 00000000000000000000000000000000101 = 000000000000000000000000000000001111
OR: 000000000000000000000000000000001010 | 00000000000000000000000000000000101 = 000000000000000000000000000000001111
Shift Left 2: 10 << 2 = 40
Shift Left 2: 000000000000000000000000000000001010 << 2 = 00000000000000000000000000000000101000
Shift Right Arith: -8 >>> 1 = -4
Shift Right Arith: 11111111111111111111111111111111000 >>> 000000000000000000000000000000000001 = 11111111111111111111111111111100
Shift Right Logical: -8 >> 1 = 2147483644
Shift Right Logical: 11111111111111111111111111111111000 >> 000000000000000000000000000000000001 = 01111111111111111111111111111100
SLT: 5 < 10 ? 1
SLTU: FFFFFFF0 < A ? 0
Default: output = 0 (expected 0)
=== END TESTBENCH ===
manbenit@NimShuriken:~/Gitlab/uvm-exercices/testbenches/02_verilator(main)$
```



veripython

```
Verilator: Warning: 1200: $ (cdab-01001, cve-01012, btd-11240), cpu 0.010 s on 1 threads, allocated 31.932 MB
manbenit@NimShuriken:~/Gitlab/uvvm-exercices/testbenches/03_veripython(main)$ python3 ALU_tb.py
=== START TESTBENCH ===
Add: 10 + 5 = 15 (expected 15)
Sub: 10 - 5 = 5 (expected 5)
Prod: 10 * 5 = 50 (expected 50)
Div: 10 / 5 = 2 (expected 2)
AND: 000000000000000000000000000000001010 & 00000000000000000000000000000000101 = 00000000000000000000000000000000
XOR: 000000000000000000000000000000001010 & 00000000000000000000000000000000101 = 000000000000000000000000000000001111
OR: 000000000000000000000000000000001010 & 00000000000000000000000000000000101 = 000000000000000000000000000000001111
Shift Left 2: 10 << 2 = 40
Shift Left 2: 000000000000000000000000000000001010 << 2 = 00000000000000000000000000000000101000
Shift Right Arith: -8 >> 1 = -4
Shift Right Arith: 111111111111111111111111111111000 >> 000000000000000000000000000000000001 = 11111111111111111111111111111100
Shift Right Logical: -8 >> 1 = 2147483644
Shift Right Logical: 111111111111111111111111111111000 >> 000000000000000000000000000000000001 = 01111111111111111111111111111100
SLT: 5 < 10 ? 1
SLTU: ffffffff0 < 10 ? 0
Default: output = 0 (expected 0)
=== END TESTBENCH ===
```

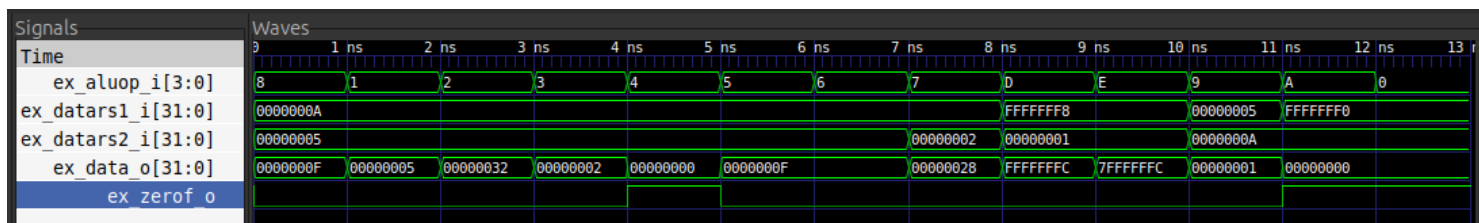


cocotb

```
FST warning: ignoring signals in previously scanned scope ALU.Adder_U1.
Add: 10 + 5 = 15 (expected 15)
Sub: 10 - 5 = 5 (expected 5)
Prod: 10 * 5 = 50 (expected 50)
Div: 10 / 5 = 2 (expected 2)
AND: 000000000000000000000000000000001010 & 00000000000000000000000000000000101 = 00000000000000000000000000000000
XOR: 000000000000000000000000000000001010 & 00000000000000000000000000000000101 = 000000000000000000000000000000001111
OR: 000000000000000000000000000000001010 & 00000000000000000000000000000000101 = 000000000000000000000000000000001111
Shift Left 2: 10 << 2 = 40
Shift Left 2: 000000000000000000000000000000001010 << 2 = 00000000000000000000000000000000101000
Shift Right Arith: -8 >> 1 = 4294967292
Shift Right Arith: 111111111111111111111111111111000 >> 000000000000000000000000000000000001 = 11111111111111111111111111111100
Shift Right Logical: -8 >> 1 = 2147483644
Shift Right Logical: 111111111111111111111111111111000 >> 000000000000000000000000000000000001 = 01111111111111111111111111111100
SLT: 5 < 10 ? 1
SLTU: -16 < 10 ? 0
Default: output = 0 (expected 0)
=== END TESTBENCH ===
13.00ns INFO cocotb.regression
13.00ns INFO cocotb.regression

alu_test passed
*****
** TEST                                STATUS SIM TIME (ns) REAL TIME (s) RATIO (ns/s) **
*****
** ALU tb.alu_test                     PASS                      13.00          0.00    5810.35 **
*****
** TESTS=1 PASS=1 FAIL=0 SKIP=0                13.00          0.01    2302.79 **
*****

make[1]: se sale del directorio '/home/manbenit/Gitlab/uvvm-exercices/testbenches/04_cocotb'
(.UVMEuv_virtualenv) manbenit@NimShuriken:~/Gitlab/uvvm-exercices/testbenches/04_cocotb(main)$
```



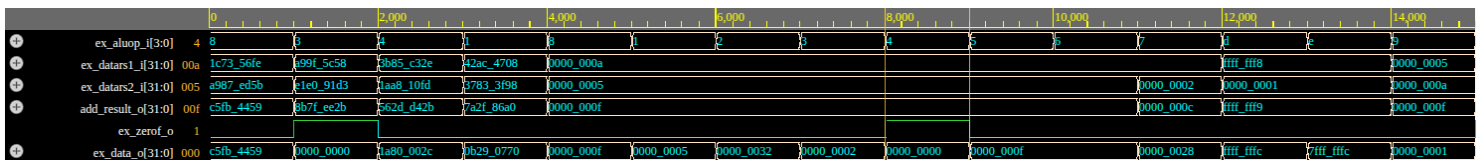
UVM nativo (usando EDAplayground)

```
UVM_INFO testbench.sv(255) @ 16000: uvm_test_top.env.agent.monitor [MON]
SLTU: ffffffff0 < 0000000a ?      0
UVM_INFO testbench.sv(353) @ 16000: uvm_test_top.env.scb [SCB] PASS:
UVM_INFO testbench.sv(196) @ 16000: uvm_test_top.env.agent.seqr@seq_directed [SEQ] === END TESTBENCH ===
UVM_INFO /apps/vcsmx/vcs/U-2023.03-SP2/etc/uvm-1.2/src/base/uvm_objection.svh(1276) @ 16000: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'e
UVM_INFO testbench.sv(412) @ 16000: uvm_test_top.env.cov [COV] Cobertura funcional alcanzada: 86.11%
UVM_INFO testbench.sv(413) @ 16000: uvm_test_top.env.cov [COV] Número total de transacciones observadas: 17
UVM_INFO /apps/vcsmx/vcs/U-2023.03-SP2/etc/uvm-1.2/src/base/uvm_report_server.svh(904) @ 16000: reporter [UVM/REPORT/SERVER]
--- UVM Report Summary ---
```

```

** Report counts by severity
UVM_INFO : 42
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[COV] 2
[MON] 17

```



Note: To revert to EPWave opening in a new browser window, set that option on your profile page.

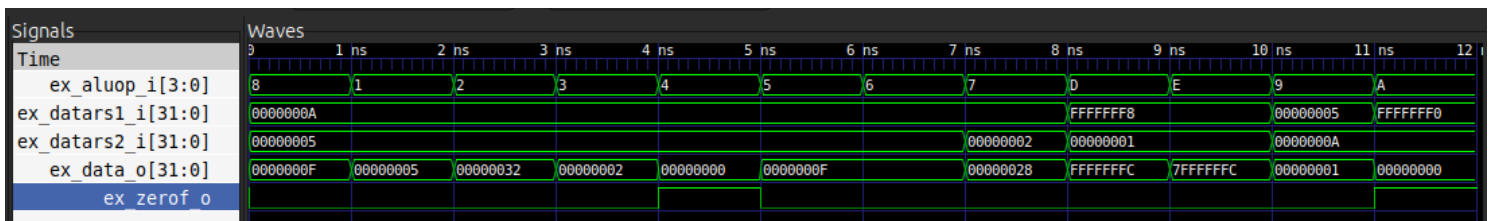
PyUVM

```

Default: output = 0x1
12.00ns INFO ..tbenchos/06_pyuvvm/ALU_tb.py(217) [uvvm_test_top.env.agent.monitor]: [MON] valor del resultado: Transaction response -> {
    "request": {
        "ex_aluop_i": "0xa",
        "ex_datars1_i": "0xffffffff0",
        "ex_datars2_i": "0xa"
    },
    "response": {
        "ex_zeroof_o": "0x1",
        "ex_data_o": "0x0"
    }
}

Default: output = 0x0
12.00ns INFO ..tbenchos/06_pyuvvm/ALU_tb.py(349) [uvvm_test_top.env.cov]: [COV] Cobertura aluop: 0.00%
12.00ns INFO ..tbenchos/06_pyuvvm/ALU_tb.py(350) [uvvm_test_top.env.cov]: [COV] Cobertura zero: 0.00%
12.00ns INFO ..tbenchos/06_pyuvvm/ALU_tb.py(351) [uvvm_test_top.env.cov]: [COV] Cobertura cruzada: 50.00%
12.00ns INFO ..tbenchos/06_pyuvvm/ALU_tb.py(352) [uvvm_test_top.env.cov]: [COV] Número total de transacciones observadas: 12
12.00ns INFO cocotb.regression test passed
12.00ns INFO cocotb.regression
*****
** TEST STATUS SIM TIME (ns) REAL TIME (s) RATIO (ns/s) **
*****
** ALU_tb.test PASS 12.00 0.01 2162.56 **
*****
** TESTS=1 PASS=1 FAIL=0 SKIP=0 12.00 0.02 758.89 **
*****

```



UVMEnv

[illegible]